

Tomato Early Blight: Identification and Management (Sample Extension Guide)

Early blight is a common fungal disease of tomato caused by *Alternaria solani*. Unlike late blight, early blight tends to develop more gradually and is often first observed on older, lower leaves of the plant.

Symptoms include small, dark brown to black spots with a characteristic pattern of concentric rings, giving lesions a target-like or bulls-eye appearance. A yellow halo often surrounds each lesion. As the disease progresses, affected leaves turn yellow, wither, and drop prematurely, which can lead to sunscald on exposed fruit.

Stem lesions appear as dark, sunken areas, often near the soil line on young seedlings, sometimes called collar rot. Fruit can also be infected near the stem end, showing dark, leathery, concentrically ringed lesions.

The disease is favored by warm temperatures between 75 and 85 degrees Fahrenheit combined with periods of high humidity or leaf wetness. It survives on infected plant debris, in soil, and can be seed-borne.

Management strategies include crop rotation with non-host crops for at least two to three years, removal of infected plant debris after harvest, staking or caging plants to improve air circulation, mulching to reduce soil splash onto lower leaves, and maintaining balanced plant nutrition since stressed plants are more susceptible.

Fungicide programs, where warranted, should begin at first symptom appearance or according to a recognized disease-forecasting model, and should be rotated among different fungicide classes to reduce the risk of resistance development. Always follow label directions and consult local extension services for approved products in your region.